PROJECT PLANNING

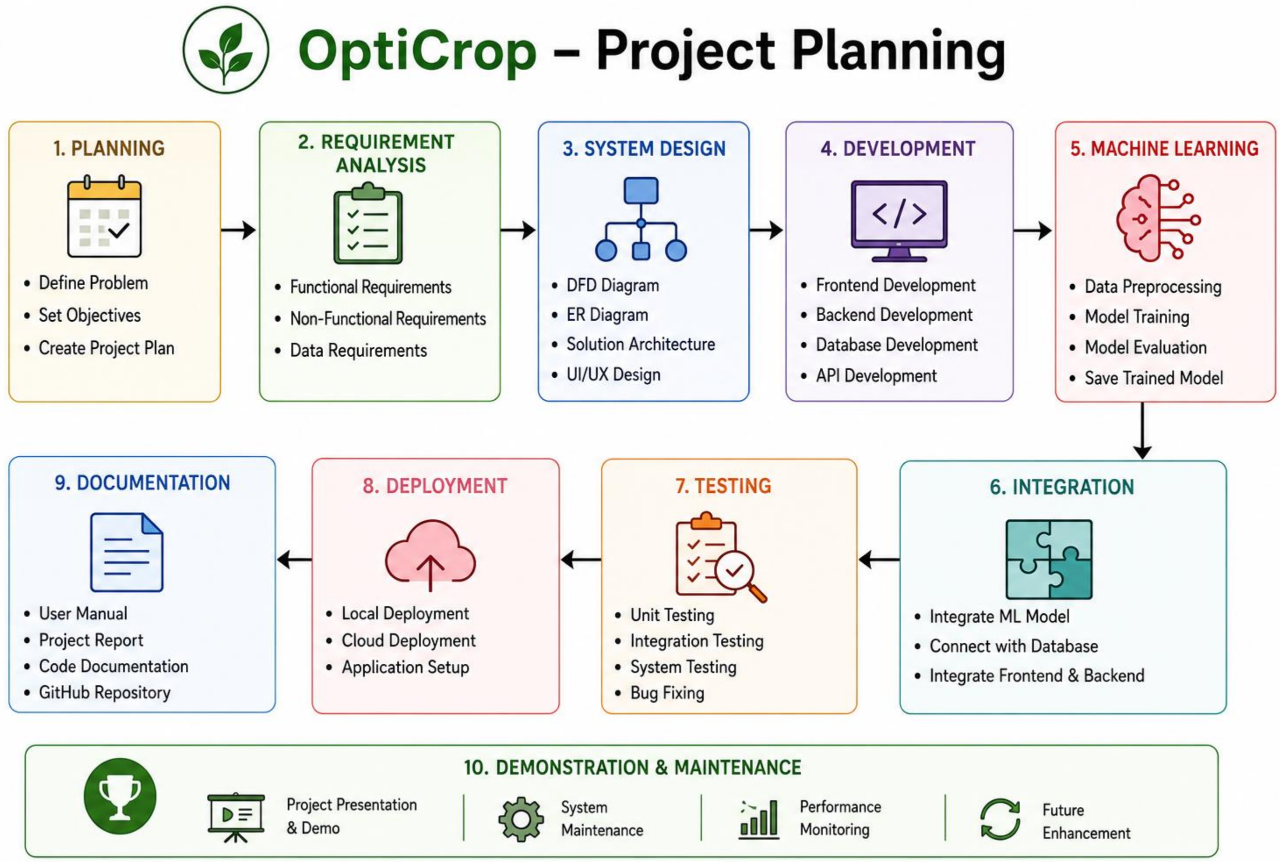
Date	27 JUNE 2026
Team ID	SWTID-2026-6540
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	

PROJECT OVERVIEW:

The goal of **opticrop** is to develop a machine learning-based web application that assists farmers in making informed agricultural decisions. The system analyzes soil, weather, and historical agricultural data to recommend suitable crops, predict crop yield, identify production patterns, and optimize resource utilization.

Project Planning Phases

Phase	Activities	Deliverables
Phase 1	Problem Identification	Problem Statement, Objectives
Phase 2	Requirement Analysis	Functional & Non-Functional Requirements
Phase 3	Data Collection	Agricultural Dataset, Weather & Soil Data
Phase 4	System Design	Architecture Diagram, Data Flow Diagram, ER Diagram
Phase 5	Development	Frontend, Backend, Database Integration
Phase 6	Machine Learning	Data Preprocessing, Model Training, Prediction
Phase 7	Testing	Functional Testing, Integration Testing
Phase 8	Deployment	Flask Application Deployment
Phase 9	Documentation	GitHub Repository, Project Report
Phase 10	Demonstration	Final Presentation and Project Demo



OptiCrop – Project Planning
1. PLANNING
Define Problem
Set Objectives
Create Project Plan
2. REQUIREMENT ANALYSIS
Functional Requirements
Non-Functional Requirements
Data Requirements
3. SYSTEM DESIGN
DFD Diagram
ER Diagram
Solution Architecture
UI/UX Design
4. DEVELOPMENT
Frontend Development
Backend Development
Database Development
API Development
5. MACHINE LEARNING
Data Preprocessing
Model Training
Model Evaluation
Save Trained Model
6. INTEGRATION
Integrate ML Model
Connect with Database
Integrate Frontend & Backend
7. TESTING
Unit Testing
Integration Testing
System Testing
Bug Fixing
8. DEPLOYMENT
Local Deployment
Cloud Deployment
Application Setup
9. DOCUMENTATION
User Manual
Project Report
Code Documentation
GitHub Repository
10. DEMONSTRATION & MAINTENANCE
Project Presentation & Demo
System Maintenance
Performance Monitoring
Future Enhancement